

Hypothesis and Sampling

Chapter 4

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What is hypothesis?

- Hypothesis, simply means a mere assumption or some supposition to be proved or disproved.
- A research hypothesis is a predictive statement, capable of being tested by scientific methods, that relates an independent variable to some dependent variable.
 - Eg: “Students who receive counselling will show a greater increase in creativity than students not receiving counselling”
- A hypothesis states what we are looking for and it is a proposition which can be put to a test to determine its validity
- . It is an informed/ educated guess which describes a possible relationship between two or more phenomenon and variables

Nature & Characteristics of a hypothesis

- Hypothesis should be clear and precise
- Should be capable of being tested.
- Should state relationship between variables
- Should be limited in scope and must be specific
- Should be stated in simple terms, as far as possible

- Should be consistent with most known facts.
- Should be amenable to testing within a reasonable time.
- Must explain the facts that give rise to the need for explanation.

Significance of hypothesis

- It offers explanations for the relationships between those variables that can be empirically tested.
- It furnishes proof that the researcher has enough background knowledge to enable the person to make suggestions in order to extend existing knowledge.
- It gives direction to an investigation.
- It structures the next phase in the investigation and therefore furnishes continuity to the examination of the problem.
- It is a powerful tool in research process to achieve dependable knowledge

Importance of hypothesis in Research

- It facilitates the extension of knowledge in an area.
- It enables the researcher to relate logically known facts to intelligent guesses about unknown conditions.
- It provides direction to the research.
- It provides a basis to the researcher for selecting sample and statistical techniques.
- It provides the basis for reporting the conclusions of the study.

Types of hypothesis

- Research hypothesis □ : defines a relationship between variables and indicates the nature of the relationship.
 - Eg : If A is valid, B follows. A student who prepares well for exam will have less of test anxiety.
 - In above eg, The researcher relates two variables like test anxiety and study habits.

- Null hypothesis □ A null hypothesis challenges the assertion of a research hypothesis and denies it altogether.
 - Null hypothesis states that even wherein the alternate hypothesis appears true; it is far due to mere threat.
 - The researcher should prove the alternate hypothesis true, thereby rejecting the Null hypothesis.

- The null hypothesis can be further categorized in the following two categories:
 - Non directional null hypothesis: This type will test for differences or relationships.
 - Examples
 - There is no difference between two groups on variable x.
 - There is no difference among three or more groups on variable x.
 - There is no relationship between variable x and variable y

- Directional null hypothesis: This type can predict the direction of the differences.
- Example:
 - Group B has a higher mean score than Group A.
 - Group B has a higher mean score than Group A and Group C.
 - There is no positive relationship between the variables x and y .

Sources of Hypothesis

- Following are the sources of hypothesis:
 - The resemblance between the phenomenon.
 - Observations from past studies, present-day experiences and from the competitors.
 - Scientific theories.
 - General patterns that influence the thinking process of people.

What is Sampling

- Defined as the selection of some part of an aggregate or totality on the basis of which a judgment or inference about the aggregate or totality is made.
- In other words, it is the process of obtaining information about an entire population by examining only a part of it. In most of the research work and surveys, the usual approach happens to be to make generalizations or to draw inferences based on samples about the parameters of population from which the samples are taken.
- Sampling done on the assumption that the sample data will enable him to estimate the population parameters.

- The items so selected constitute what is technically called a sample, their selection process or technique is called sample design and the survey conducted on the basis of sample is described as sample survey.
- Sample should be truly representative of population characteristics without any bias so that it may result in valid and reliable conclusions.

- Need for sampling :
 - Sampling can save time and money.
 - Sampling may enable more accurate measurements for a sample study is generally conducted by trained and experienced investigators.
 - Sampling remains the only way when population contains infinitely many members.
 - Sampling usually enables to estimate the sampling errors and, thus, assists in obtaining information concerning some characteristic of the population

Characteristics of a good sample

- Representativeness
- Absence of sampling error
- Economically viable
- Generalized and applicable
- Goal oriented
- Proportional
- Randomly selected
- Practical

Types of sampling methods

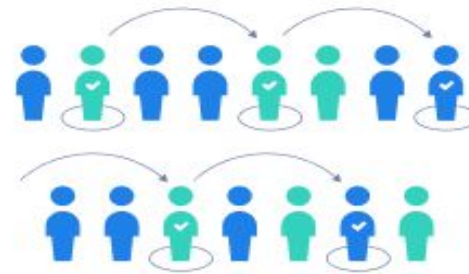
- Two primary types of sampling methods
 - **Probability sampling methods** □ involves random selection, allowing you to make strong statistical inferences about the whole group.
 - **Non-probability sampling methods** □ involves non-random selection based on convenience or other criteria, allowing you to easily collect data.

- Probability sampling ☐
 - every member of the population has a chance of being selected. It is mainly used in Quantitative research.
 - If we want to produce results that are representative of the whole population, probability sampling techniques are the most valid choice.
 - Four types ☐ Simple, Systematic, Stratified, Cluster

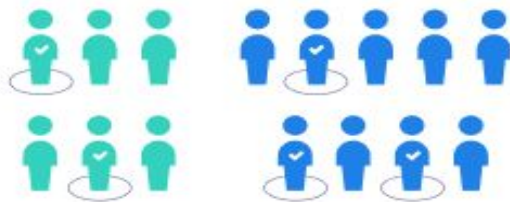
Simple random sample



Systematic sample



Stratified sample



Cluster sample



- Eg Simple random Sampling
 - To select a simple random sample of 1000 employees of a social media marketing company. We assign a number to every employee in the company database from 1 to 1000, and use a random number generator to select 100 numbers
- Eg of Systematic sampling
 - All employees of the company are listed in alphabetical order. From the first 10 numbers, we randomly select a starting point: number 6. From number 6 onwards, every 10th person on the list is selected (6, 16, 26, 36, and so on), and we end up with a sample of 100 people.

- Eg of Stratified Sampling

- The company has 800 female employees and 200 male employees. We want to ensure that the sample reflects the gender balance of the company, so we sort the population into two strata based on gender. Then use random sampling on each group, selecting 80 women and 20 men, which gives us a representative sample of 100 people.

- Eg of Cluster Sampling
 - The company has offices in 10 cities across the country (all with roughly the same number of employees in similar roles). You don't have the capacity to travel to every office to collect your data, so you use random sampling to select 3 offices – these are your clusters.

Non-probabilistic sampling

- Often associated with case study research design and qualitative research.
- Case studies tend to focus on small samples and are intended to examine a real life phenomenon, not to make statistical inferences in relation to the wider population (Yin, 2003).

- Quota Sampling :

- A non random sampling technique in which participants are chosen on the basis of predetermined characteristics so that the total sample will have the same distribution of characteristics as the wider population.
- Eg: Research related to TV violence study where the population is all middle-school children in the Conway, Arkansas, school district (approximately 1000 students)
 - For Quota sampling, the researcher selects, first 20 sixth-grade boys, first 20 sixth grade girls , first 20 seventh grade boys, first 20 seventh grade girls , first 20 eighth grade boys and first 20 eighth grade girls.

- Snowball sampling
 - A non random method that involves building a participant pool by having one potential lead refer the researcher to other potential participants.
 - It's also known as chain-referral or network sampling



- This approach is most applicable in small populations that are difficult to access due to their closed nature, e.g. secret societies and inaccessible professions.
- Can estimate rare characteristics but Time-consuming
- Eg:
 - Many researchers in the medical field use snowball sampling to contact people who have rare or less-researched diseases.
 - it might be challenging to survey the top 1% of car owners who own premium luxury cars unless you ask sources to share referrals with you.

- Convenience sampling
 - Convenience sampling is selecting participants because they are often readily and easily available.
 - Problems like Selection bias, sample not representative, not recommended by descriptive or casual research, can come up
 - Eg : Researcher know one of the middle school teachers, and the teacher volunteers her 35 students for the study.

Sample design

- While developing a sampling design, the researcher must pay attention to the following points:
 - **Type of universe:**
 - The first step in developing any sample design is to clearly define the set of objects, technically called the Universe, to be studied.
 - The universe can be finite or infinite
 - The population of a city, the number of workers in a factory and the like are examples of finite universes,
 - The number of stars in the sky, listeners of a specific radio programme etc. are examples of infinite universes.

- **Sampling unit:** A decision has to be taken concerning a sampling unit before selecting sample.
 - Sampling unit may be a geographical one such as state, district, village, etc., or a construction unit such as house, flat, etc., or it may be a social unit such as family, club, school, etc., or it may be an individual.
- **Source list:** Also known as ‘sampling frame’ from which sample is to be drawn.
 - It contains the names of all items of a universe (in case of finite universe only). If source list is not available, researcher has to prepare it.

- **Size of sample:** This refers to the number of items to be selected from the universe to constitute a sample. This is a major problem before a researcher. The size of sample should neither be excessively large, nor too small. It should be optimum
- **Parameters of interest:** In determining the sample design, one must consider the question of the specific population parameters which are of interest.
 - For instance, we may be interested in estimating the proportion of persons with some characteristic in the population, or we may be interested in knowing some average or the other measure concerning the population

- **Budgetary constraint:** Cost considerations, from practical point of view, have a major impact upon decisions relating to not only the size of the sample but also to the type of sample. This fact can even lead to the use of a non-probability sample.
- **Sampling procedure:** Finally, the researcher must decide the type of sample he will use i.e., he must decide about the technique to be used in selecting the items for the sample.

Choice of sampling technique

- 6 stages of choosing sampling technique:
- **Stage 1: Clearly Define Target Population**
 - The first stage in the sampling process is to clearly define target population. Population is commonly related to a group or number of elements that researcher plans to study among

Stage2: Select Sampling Frame

- A sample frame is drawn from the identified population/target population.
- A sampling frame is a list of the actual cases from which sample will be drawn.
- Thus, the sampling frame must be representative of the population. In brief, sampling frame is a list of all the items in the research population

Stage 3: Choose Sampling Technique

- Taking a subset from chosen sampling frame or entire population is called sampling.
- Sampling can be used to make inference about a population or to make generalization in relation to existing theory. In essence, this depends on choice of sampling technique.

- **Stage 4: Determine Sample Size**

- In order to generalize from a random sample and avoid sampling errors or biases, a random sample needs to be of adequate size.
- It is important that Size is not proportion of the research population that gets sampled, but
 - the absolute size of the sample selected relative to the complexity of the population,
 - the aims of the researcher
 - the kinds of statistical manipulation that will be used in data analysis.
- Larger sample sizes reduce sampling error but at a decreasing rate.

- **Stage 5: Determine Sampling technique**

- Below are some steps expert researchers follow to decide the best sampling method.

- Jot down the research goals. Generally, it must be a combination of cost, precision, or accuracy.
 - Identify the effective sampling techniques that might potentially achieve the research goals.
 - Test each of these methods and examine whether they help achieve your goal.
 - Select the method that works best for the research.

- **Comparison between Probability Sampling Methods & Non-Probability Sampling Methods :**
 - **Probability Sampling** is a sampling technique in which samples from a larger population are chosen using a method based on the theory of probability.
 - **Non-probability sampling** is a sampling technique in which the researcher selects samples based on the researcher's subjective judgment rather than random
 - **Population selection** The population is selected randomly. The population is selected arbitrarily.
 - **Nature** The research is conclusive. The research is exploratory.
 - **Sample** Since there is a method for deciding the sample, the population demographics are conclusively represented. Since the sampling method is arbitrary, the population demographics representation is almost always skewed

- **Time Taken** : Takes longer to conduct since the research design defines the selection parameters before the market research study begins. This type of sampling method is quick since neither the sample nor the selection criteria of the sample are undefined.
- **Results** : This type of sampling is entirely unbiased; hence, the results are also conclusive. This type of sampling is entirely biased, and hence the results are biased, too, rendering the research speculative.
- **Hypothesis** : In probability sampling, there is an underlying hypothesis before the study begins, and this method aims to prove the hypothesis. In non-probability sampling, the hypothesis is derived after conducting the research study.

Examples in CS

- **Software Engineering & Code Quality**
 - **Problem:** Does code review by AI-based tools reduce defect density compared to human-only reviews?
 - **Hypothesis:** AI-assisted code reviews detect more defects than traditional reviews.
 - **Sampling:** Select a random set of projects/repos from GitHub across different domains.

- **Cybersecurity (Intrusion Detection Systems)**

- **Problem:** Does using deep learning improve detection accuracy of network intrusions compared to traditional rule-based systems?
- **Hypothesis:** Deep learning models have significantly higher detection rates than classical methods.
- **Sampling:** Draw samples of network traffic logs from datasets like NSL-KDD, UNSW-NB15.

- **Human–Computer Interaction (HCI)**
 - **Problem:** Do dark mode interfaces reduce eye strain compared to light mode in extended computer use?
 - **Hypothesis:** Participants using dark mode will report less eye strain and fatigue.
 - **Sampling:** Randomly select participants from student/employee populations.

- **Natural Language Processing (NLP)**

- **Problem:** Does fine-tuning large language models with domain-specific data improve sentiment analysis accuracy?
- **Hypothesis:** Fine-tuned models outperform general-purpose models on domain-specific tasks.
- **Sampling:** Randomly sample reviews from e-commerce, healthcare, or finance datasets.

- **Cloud Computing & Performance**

- **Problem:** Does container-based deployment (Docker, Kubernetes) provide better scalability than virtual machines for web applications?
- **Hypothesis:** Container-based systems will have lower response times under high load.
- **Sampling:** Collect performance data by running applications with sampled traffic loads.

- **Education Technology (EdTech / e-Learning)**
 - **Problem:** Does gamification in learning platforms increase student engagement compared to non-gamified systems?
 - **Hypothesis:** Students using gamified platforms show higher completion rates.
 - **Sampling:** Randomly select student groups across courses/universities.

- In each of these problems:
 - **Hypotheses** are clear statements to be tested.
 - **Sampling techniques** (random, stratified, cluster sampling, etc.) are used to select datasets, participants, or system performance trials.